

where

$$A_n = \frac{8b}{n^2\pi^2} \sin \frac{n\pi}{2} \quad n = 1, 3, 5, \dots,$$

and

$$A_n = 0 \quad n = 2, 4, 6, \dots$$

The factor $\sin(n\pi/2)$ is included to give the sign of the odd n terms and to delete all the even n terms. Finally, the waveform of the plucked string is

$$y(x, t) = \sum_{n=\text{odd}}^{\infty} \left(\frac{8b}{n^2\pi^2} \sin \frac{n\pi}{2} \right) \cos \omega_n t \sin \frac{n\pi x}{L}.$$

Exercise 13.5

Equation (13.6) is a Fourier series. Plotting the terms in this series, we find the integer n is equal to the number of *loops* in the standing wave when only the y_n term is excited. The various modes, $n = 1, 2, 3, \dots$ are called the first, second, third $\dots$ harmonics. Plot the first four harmonics for a string of length L .

13.4 Traveling Waves

In the previous section we considered a standing wave. Now consider a traveling wave, such as a pulse traveling down an infinite string (or an electromagnetic wave propagating in empty space). Imagine a very long (perhaps infinite) string that is plucked at some point, or perhaps shaken at one end. Your experience with ropes and slinkies tells you that a pulse will move down the string, as pictured in [Figure 13.5](#). If there is no loss of energy, the shape will be repeated at a later time at a further position, as shown in the lower panel of the figure. Suppose the top panel gives the waveform at time $t = 0$, so the plot is $y = y(x, 0)$. The bottom panel shows the wave at a later time t . But $y(x, t) = y(x - vt, 0)$, that is, the displacement at point x at time t is the same as the displacement at $x - vt$ at the

earlier time. Note that during the time interval t , the wave has traveled a distance vt . Therefore, v is the speed at which the waveform propagates. This is called the *phase speed*. (Recall that we identified v with the quantity $\sqrt{F/\rho}$.)

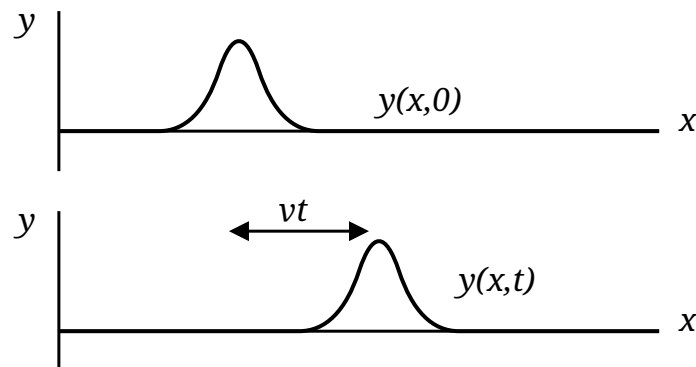


Figure 13.5 A pulse in a string moving to the right. It does not change shape, so the displacement obeys the condition $y(x, t) = y(x + dx, t + dt)$.

The fact that the displacement at time t is the same as the displacement at $x - vt$ at an earlier time suggests that the waveform for a wave traveling at speed v will have the functional form

$$y(x, t) = f(x - vt).$$

In [Figure 13.5](#), the lower panel represents a later time, so the wave is moving toward the right, that is, toward positive x . A wave traveling in the opposite direction would have the functional form

$$y(x, t) = g(x + vt).$$

For example, a sinusoidal traveling wave moving toward the right can be represented as

$$y(x, t) = A \sin k(x - vt).$$

Here I introduced k simply as a constant to make the argument of the sine dimensionless; however, if you think about it, you will realize that k is actually the wave number, so the expression becomes

$$y(x, t) = A \sin(kx - \omega t).$$

You might plot this expression for a few nearby values of time and verify for yourself that this indeed is a representation of a traveling sine wave moving in the positive x -direction.

Although the expression

$$y = f(x - vt)$$

seems a reasonable way to describe a traveling wave, it remains to be shown that it satisfies the wave equation for any function f that depends on x and t in the combination $x - vt$. For example, a number of different possible forms of f would be

$$y(x, t) = A \cos k(x - vt)$$

$$y(x, t) = Ae^{ik(x-vt)}$$

$$y(x, t) = A \sin(k^2(x - vt)^2).$$

The assumption that any function of $(x - vt)$ satisfies the wave equation can be proved by plugging $y = f(x - vt)$ into the equation and noting that if $u = x - vt$, then $\frac{\partial f}{\partial t} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial t} = -v \frac{\partial f}{\partial u}$, so

$$\begin{aligned} \frac{\partial^2 y}{\partial t^2} &= \frac{\partial}{\partial t} \left[\frac{\partial f(u)}{\partial t} \right] = -v \frac{\partial}{\partial t} \left[\frac{\partial f}{\partial u} \right] \\ &= -v \left[\left(\frac{\partial}{\partial u} \frac{\partial f}{\partial u} \right) \frac{\partial u}{\partial t} \right] = +v^2 \frac{\partial^2 f}{\partial u^2}, \end{aligned}$$

and since $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial x} = \frac{\partial f}{\partial u} (1)$, we can write

$$\begin{aligned} v^2 \frac{\partial^2 y}{\partial x^2} &= v^2 \frac{\partial}{\partial x} \left[\frac{\partial f}{\partial x} \right] = v^2 \frac{\partial}{\partial x} \left[\frac{\partial f}{\partial u} \right] \\ &= v^2 \frac{\partial}{\partial u} \left[\frac{\partial f}{\partial u} \right] = v^2 \frac{\partial^2 f}{\partial u^2}. \end{aligned}$$

The right-hand sides of the two expressions are equal, proving the wave equation is satisfied for *any* function of $x - vt$.

A particularly useful function of $x - vt$ that is often used to represent a traveling wave is

$$y(x, t) = Ae^{ik(x-vt)} = Ae^{i(kx-\omega t)}.$$

Note that this can be expressed in terms of sines and cosines using the Euler relation:

$$y(x, t) = Ae^{i(kx-\omega t)} = A[\cos(kx - \omega t) + i \sin(kx - \omega t)].$$

Although this expression satisfies the wave equation, it is not obvious how to interpret the imaginary part. Therefore, we avoid the problem by writing

$$y(x, t) = \text{Re} \left[Ae^{i(kx-\omega t)} \right],$$

that is, only taking the real part of the expression. This formulation is particularly useful when studying electromagnetic waves.

Exercise 13.6

Argue that the constant k introduced above as a proportionality constant must actually be the wave number.

Exercise 13.7

Plot $y = A \sin(kx - \omega t)$ for four nearby values of time to show that this indeed represent a wave traveling to the right. (Select any reasonable values for the parameters, but make sure that the four values of t are less than a quarter of the period.)

13.5 Standing Waves as a Special Case of Traveling Waves

You have just seen that the solution to the wave equation is any function of $(kx \pm \omega t)$. But the standing wave of [Figure 13.1](#) that was considered earlier was also a solution of the wave equation and it had the form

$$y(x, t) = A \sin kx \cos \omega t + B \sin kx \sin \omega t. \quad (13.7)$$

This expression is a very slightly modified version of [Equation \(13.4\)](#). It does not

look much like a function of $(kx \pm \omega t)$! Are the two expressions equivalent? The answer is yes, and the underlying reason is that a standing wave can be considered to be the sum of two traveling waves moving in opposite directions. We now show that this is indeed true.

To be a bit less general but to make the analysis easier to follow, assume the traveling waves are sinusoidal. The general form of two sinusoidal waves traveling in opposite directions is a combination of sine and cosine terms, thus,

$$y(x, t) = A \sin(kx - \omega t) + B \sin(kx + \omega t) + C \cos(kx - \omega t) + D \cos(kx + \omega t).$$

The boundary conditions require that $y = 0$ at $x = 0$ and at $x = L$. If $x = 0$

$$\begin{aligned} 0 &= -A \sin \omega t + B \sin \omega t + C \cos \omega t + D \cos \omega t, \\ &= (B - A) \sin \omega t + (C + D) \cos \omega t. \end{aligned}$$

This condition holds at all times if

$$\begin{aligned} B &= A, \\ C &= -D. \end{aligned}$$

and the solution reduces to

$$y = A[\sin(kx - \omega t) + \sin(kx + \omega t)] + C[\cos(kx - \omega t) - \cos(kx + \omega t)].$$

Imposing the second boundary condition ($y = 0$ at $x = L$) gives

$$0 = A[\sin(kL - \omega t) + \sin(kL + \omega t)] + C[\cos(kL - \omega t) - \cos(kL + \omega t)].$$

This expression is valid at all times, including $t = 0$, in which case it is easy to appreciate that k is equal to any of the values

$$k = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots$$

(Recall that the standing wave satisfied this condition.)

The sum of the two traveling waves (with $y = 0$ at $x = 0, L$) is, therefore,

$$y(x, t) = A[\sin(kx - \omega t) + \sin(kx + \omega t)] + C[\cos(kx - \omega t) - \cos(kx + \omega t)],$$

with $k = n\pi/L$ and $\omega = kv = n\pi v/L$.

Using the trigonometric identities for the sums and differences of angles,⁶

$$y(x, t) = [2A \sin(kx)] \cos \omega t + [2C \sin(kx)] \sin \omega t.$$

Recalling that the standing wave was given by

$$y(x, t) = A \sin(kx) \cos \omega t + B \sin(kx) \sin \omega t,$$

we see that the two expressions are identical except for the notation of the coefficients. Thus, we have shown that the standing wave can be considered the superposition of two traveling waves moving in opposite directions.

Exercise 13.8

Show that $y = 0$ when $x = L$.

13.6 Energy

Let us determine the energy in a wave. As an example, consider a standing wave in a string of length L .

The kinetic energy of the mass element ρdx as it oscillates up and down is

$$dT = \frac{1}{2} \rho dx \left(\frac{dy}{dt} \right)^2.$$

To determine the potential energy note that the element of unstretched length dx has stretched length ds , so the displacement produces a net stretch in the string of $ds - dx$. (See [Figure 13.2](#).) The tension in the string is F , so the work done to stretch the string (which is equal to the increase in its potential energy) is

$$dV = F(ds - dx).$$

⁶ These are:

$$\sin \alpha + \sin \beta = 2 \sin \frac{1}{2}(\alpha + \beta) \cos \frac{1}{2}(\alpha - \beta)$$

$$\cos \alpha - \cos \beta = -2 \sin \frac{1}{2}(\alpha + \beta) \sin \frac{1}{2}(\alpha - \beta).$$

But

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx,$$

so

$$dV = F \left[\sqrt{1 + \left(\frac{dy}{dx}\right)^2} - 1 \right] dx.$$

The binomial expansion of the square root leads to

$$dV = F dx \left[1 + \frac{1}{2} \left(\frac{dy}{dx}\right)^2 + \cdots - 1 \right] \doteq F \frac{1}{2} \left(\frac{dy}{dx}\right)^2 dx.$$

Consequently, the total energy in a vibrating string of length L is

$$E = T + V = \int_0^L (dT + dV) = \int_0^L \left[\frac{1}{2} \rho \left(\frac{\partial y}{\partial t}\right)^2 + \frac{1}{2} F \left(\frac{\partial y}{\partial x}\right)^2 \right] dx.$$

For example, if the wave is described by

$$y(x, t) = A \sin \frac{\pi x}{L} \cos \omega t,$$

then

$$\frac{\partial y}{\partial t} = -\omega A \sin \frac{\pi x}{L} \sin \omega t,$$

and

$$\frac{\partial y}{\partial x} = \frac{\pi}{L} A \cos \frac{\pi x}{L} \cos \omega t.$$

Consequently,

$$\begin{aligned} E &= \int_0^L \left[\frac{1}{2} \rho \omega^2 A^2 \left(\sin^2 \frac{\pi x}{L} \sin^2 \omega t \right) + \frac{1}{2} F \frac{\pi^2}{L^2} A^2 \left(\cos^2 \frac{\pi x}{L} \cos^2 \omega t \right) \right] dx, \\ &= \frac{1}{2} A^2 \left[\rho \omega^2 \sin^2 \omega t \int_0^L \sin^2 \frac{\pi x}{L} dx \right] + \frac{1}{2} A^2 \left[F \frac{\pi^2}{L^2} \cos^2 \omega t \int_0^L \cos^2 \frac{\pi x}{L} dx \right], \\ &= \frac{1}{2} A^2 \frac{L}{2} \left[\rho \omega^2 \sin^2 \omega t + F \frac{\pi^2}{L^2} \cos^2 \omega t \right]. \end{aligned}$$

Now recall that $F/\rho = v^2$ and $\omega = \pi v/L$, so $F\pi^2/L^2 = \rho v^2 \pi^2 / (\pi^2 v^2 / \omega^2) = \rho \omega^2$ and

$$\begin{aligned} E &= \frac{1}{2} A^2 \frac{L}{2} \rho \omega^2 (\sin^2 \omega t + \cos^2 \omega t) \\ &= A^2 \rho \omega^2 \frac{L}{4}. \end{aligned}$$

Note that the energy is proportional to the amplitude squared.

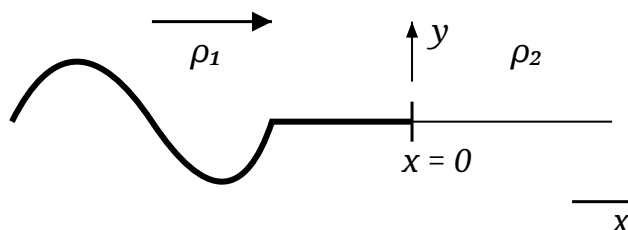


Figure 13.6 Two strings of different linear densities are connected at $x = 0$.

13.6.1 Energy Flow

Consider a wave passing from one medium into another. An example is the sunlight incident on the window of your room. As you know, some of the incident light will be transmitted and some will be reflected. Reflection is a characteristic property of any kind of wave when it is incident upon the interface of two materials in which the wave travels at different speeds. For the light wave, glass has a larger index of refraction than air, and the wave travels more slowly in glass. For a wave in a string, the speed depends on the mass per unit length of the string (ρ).

An analogy to a light wave incident on glass would be a traveling wave in a string of linear density ρ_1 incident on a second string of different linear density ρ_2 . This is illustrated in [Figure 13.6](#).

Imagine a wave coming in from the left (from $x = -\infty$). Call this the *incident* wave. At the junction of the two strings (at $x = 0$), the incident wave is split into two waves, the *reflected* wave that moves back toward $x = -\infty$, and the *transmitted*

wave that continues on toward $x = +\infty$. Let us denote these three waves by y_i, y_r , and y_t . These are all traveling waves, so they can be represented as follows:

$$\begin{aligned} y_i &= A_i \cos(k_1 x - \omega t), \\ y_r &= A_r \cos(k_1 x + \omega t), \\ y_t &= A_t \cos(k_2 x - \omega t). \end{aligned} \quad (13.8)$$

Note the sign on the ωt term in the expression for y_r and the subscript on k for y_t .

The speed of the wave is different in the two mediums. Because $v = \omega/k$, a change in the speed could be due to a change in either ω or k (or both). But it turns out that the waves have the same value of ω . You may wonder why k changes but ω stays the same. The reason is that the frequency of the wave is determined by the physical mechanism that is generating the wave. If you are standing at one end of the string and shaking it up and down, you are controlling the frequency of the wave by how quickly (or slowly) you move the end of the rope. This you can control. However, you have no control over the wavelength. The wavelength depends on the properties of the medium. When the properties of the medium change the wavelength changes, and so the wave number $k = 2\pi/\lambda$ also changes.

The energy of a wave is proportional to the square of its amplitude. An interesting and important question is how much energy is reflected at an interface and how much is transmitted into the second medium. The question can be answered by determining the amplitudes of the reflected and transmitted waves relative to the incident amplitude. In other words, it is desirable to determine the ratios A_r/A_i and A_t/A_i . This can be done by realizing that both y and dy/dx must be continuous at the junction between the two strings. The value of y is continuous because the strings are attached to one another. The reason why dy/dx is the same on both sides of the junction is that the tension is the same on both sides. These two boundary conditions can be expressed as

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$$\begin{aligned} [y_i + y_r]_{x=0} &= [y_t]_{x=0}, \\ \left[\frac{\partial y_i}{\partial x} + \frac{\partial y_r}{\partial x} \right]_{x=0} &= \left[\frac{\partial y_t}{\partial x} \right]_{x=0}. \end{aligned}$$

Substituting the expressions from [Equations \(13.8\)](#) we obtain

$$A_i \cos(-\omega t) + A_r \cos(+\omega t) = A_t \cos(-\omega t),$$

$$A_i + A_r = A_t,$$

and

$$k_1(A_i - A_r) = k_2 A_t,$$

or

$$\frac{1}{v_1}(A_i - A_r) = \frac{1}{v_2} A_t.$$

Therefore,

$$\frac{A_r}{A_i} = \frac{v_2 - v_1}{v_1 + v_2},$$

$$\frac{A_t}{A_i} = \frac{2v_2}{v_1 + v_2}.$$

Note that, since $v = \sqrt{F/\rho}$, if the second string is heavier than the first, $\rho_2 > \rho_1$ and $v_2 < v_1$ and A_r is negative. This means the reflected wave is inverted ("upside down"). Going to the extreme case, if the second string is infinitely massive, $v_2 = 0$ and $A_t = 0$. This is equivalent to string 1 being "nailed down" at $x = 0$.

To determine the energy flow through the junction consider the rate at which work is done by a particle on one side of $x = 0$ on a particle on the other side. Assume the particle on the left (at $x = 0^-$) exerts a downward force on the particle at $x = 0^+$, given by

$$F_y = -F \sin \theta \doteq -F \tan \theta = -F \frac{\partial y}{\partial x}.$$

The rate at which work is done is the power and is given by $P = \mathbf{F} \cdot \mathbf{v}$. Therefore,

$$\begin{aligned} \left[\frac{dW}{dt} \right]_{x=0^+} &= \left[F_y \frac{\partial y}{\partial t} \right]_{x=0} \\ &= \left[\left(-F \frac{\partial y_t}{\partial x} \right) \frac{\partial y_t}{\partial t} \right]_{x=0} \\ &= -F [(k_2 A_t \sin(k_2 x - \omega t))(A_t (-\omega) \sin(k_2 x - \omega t))]_{x=0} \\ &= F A_t^2 k_2 \omega \sin^2(\omega t) = F A_t^2 (\omega^2 / v_2) \sin^2(\omega t), \end{aligned}$$

where we used $k = \omega/v$. The average of $\sin^2(\omega t)$ over one period is $1/2$. The average energy flow into string number 2 is equal to the average work done, so

$$\left\langle \frac{dE}{dt} \right\rangle_{x=0^+} = \frac{F A_t^2 \omega^2}{2v_2}.$$

An equivalent analysis for the energy leaving string number 1 gives

$$\left\langle \frac{dE}{dt} \right\rangle_{x=0^-} = \frac{F(A_i^2 - A_r^2)\omega^2}{2v_1}.$$

These expressions must be equal if there is no energy loss at the junction.

Worked Example 13.2 Obtain an expression for the reflection coefficient R defined as the ratio of the intensity of the reflected wave to the intensity of the incident wave ($R = I_r/I_i$).

Solution The intensity of a wave is defined as the energy transported across a normal unit area per unit time, that is, power per unit area. Because energy is proportional to the amplitude squared

$$\begin{aligned} R = \frac{I_r}{I_i} &= \frac{A_r^2/(\text{area} \times \text{time})}{A_i^2/(\text{area} \times \text{time})} = \frac{A_r^2}{A_i^2} = \left(\frac{v_2 - v_1}{v_2 + v_1} \right)^2 \\ &= \left(\frac{k_1 - k_2}{k_1 + k_2} \right)^2. \end{aligned}$$

Exercise 13.8

Show that if the tension is continuous at the junction, the slope dy/dx is continuous.

Exercise 13.10

Obtain the appropriate boundary condition for two strings joined by a knot of mass m . Answer:

$$F \left(\frac{\partial y}{\partial x} \Big|_{0^+} - \frac{\partial y}{\partial x} \Big|_{0^-} \right) = m \frac{\partial^2 y}{\partial t^2} \Big|_0.$$

13.7 Momentum (Optional)

The particles in a transverse wave move perpendicular to the motion of the wave. Thus, it would seem that a transverse wave would not transport momentum longitudinally. However, it is a known fact that waves in strings *do* transport momentum. This means that our assumption of purely transverse motion for a particle in a string must be wrong. There must be some longitudinal motion as well as transverse motion.⁷ This can be easily appreciated conceptually by (mentally) replacing the string with a series of particles of mass m connected by massless springs of constant k , as shown in Figure 13.7. As indicated in the figure, if one particle is plucked so that it is displaced vertically, it causes the particles on either side to be displaced both vertically *and* horizontally.

When the plucked particle is released, it sets up a (large amplitude) transverse wave as well as a (small amplitude) longitudinal wave. Rowland and Pask⁸ carried out numerical studies using

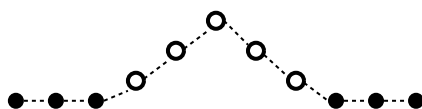


Figure 13.7 A string modeled as a series of particles connected by springs. The open circles represent the equilibrium positions of the particles. The filled circles represent their positions when a particle is displaced vertically.

⁷ Electromagnetic waves in a vacuum are transverse. The momentum density associated with these waves is $\epsilon_0 \mathbf{E} \times \mathbf{B}$, which is perpendicular to the transverse oscillations of $\mathbf{E}$ and $\mathbf{B}$. For a string, however, perfectly transverse waves would not have a longitudinal component of momentum.

⁸ David R. Rowland and Colin Pask, "The Missing Wave Momentum Mystery," *American Journal of Physics*, **67**, 378-388 (1999) and David R. Rowland, "Comment on 'What Happens to Energy and

Momentum When Two Oppositely-Moving Wave Pulses Overlap?' by N. Gauthier," *American Journal of Physics*, **72**, 1425-1429 (2004). It is interesting to note that the theory of a vibrating string was first developed by Lagrange in 1759 but there are still papers being written about it.

15 Accelerated Reference Frames

By now you have heard me say many times that Newton's laws of motion are applicable only in inertial (nonaccelerated) frames of reference. The question arises: How do we deal with motion in an *accelerated* reference frame? After all, most real reference frames are accelerating and, for many of them, the acceleration cannot be neglected. Since we live on the surface of a large rotating sphere, it is important for us to be able to solve physics problems in noninertial systems.

In this chapter you will learn how to express Newton's second law in a noninertial (accelerating) reference frame. We are, of course, particularly interested in rotating coordinate systems, which are a prime example of accelerating reference frames. Our study of these systems will introduce you to "fictitious" forces such as the centrifugal force and the Coriolis force. The Coriolis force, as you may know, is responsible for a number of important geophysical processes such as hurricanes and ocean currents. Finally, we will consider the motion of a very long simple pendulum and show that the plane of this "Foucault" pendulum precesses due to the rotation of the Earth.

15.1 A Linearly Accelerating Reference Frame

Although the most interesting noninertial reference frame for our purposes is a rotating reference frame such as the Earth, I would like to begin by looking at a reference frame that is accelerating *linearly* relative to an inertial reference frame. [Figure 15.1](#) shows a reference frame O that is at rest "with respect to the fixed stars" and a second reference frame, O' , that is accelerating at a constant acceleration $\ddot{\mathbf{r}}$ relative to O .¹ In the figure, the position of a particle of mass m with respect to the inertial frame is $\mathbf{r}_O$ and the position of the particle with respect to the accelerating

frame is $\mathbf{r}_{O'}$. The position of the origin O' with respect to O is $\mathbf{r}$. By tip to tail addition of vectors,

$$\mathbf{r}_O = \mathbf{r} + \mathbf{r}_{O'}. \quad (15.1)$$

Because O is an inertial coordinate system, Newton's second law holds in that frame and we can write

$$m\ddot{\mathbf{r}}_O = \mathbf{F}, \quad (15.2)$$

where $\mathbf{F}$ is the force acting on the particle. The acceleration of the particle relative to the accelerating frame is $\ddot{\mathbf{r}}_{O'}$. The force on the particle and the mass of the particle are assumed independent of the reference frame. (I am specifically excluding relativistic effects.)

¹ Please ignore the annoying fact that eventually O' will be moving at the speed of light!

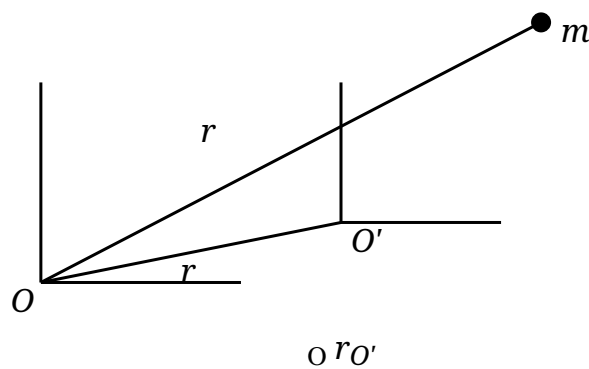


Figure 15.1 Coordinate system O' is accelerating at a constant rate $\ddot{\mathbf{r}}$ with respect to coordinate system O .

Taking the second time derivative of [Equation \(15.1\)](#) yields

$$\ddot{\mathbf{r}}_O = \ddot{\mathbf{r}}_{O'} + \ddot{\mathbf{r}},$$

where $\ddot{\mathbf{r}}$ is the acceleration of reference frame O' relative to the inertial frame O . Newton's law (15.2) then reads

$$\mathbf{F} = m\ddot{\mathbf{r}}_O = m(\ddot{\mathbf{r}}_{O'} + \ddot{\mathbf{r}}) = m\ddot{\mathbf{r}}_{O'} + m\ddot{\mathbf{r}}.$$

Therefore, from the point of view of the observer in the *accelerated reference frame*, Newton's second law has the form

$$m\ddot{\mathbf{r}}_{O'} = \mathbf{F} - m\ddot{\mathbf{r}}.$$

That is, there appears to be an *additional* force $\mathbf{f} = -m\ddot{\mathbf{r}}$ acting on the particle. If asked to determine the force on the particle, the observer in the accelerated reference frame would answer $\mathbf{F} + \mathbf{f}$.

Thus the observer in the accelerated system will believe that an additional force is acting on the particle. This is called a *fictitious force* because it is really just a mass times acceleration term which has been moved to the other side of the equation. Well-known fictitious forces include the centrifugal force and the Coriolis force.

15.2 A Rotating Coordinate Frame

Consider an ant standing still on the surface of a basketball. If the ball is rotating, the ant is moving in a circle around the axis of rotation. (The ant may *think* it is at rest. You may *think* you are sitting still in this room, but you are actually moving around in a large circle at over 600 miles per hour.) An object moving in a circle has a continuously changing velocity. Therefore, it is accelerating. The acceleration is directed towards the center of the circle. A coordinate system attached to a rotating sphere is an accelerated coordinate system. A particle at rest relative to this coordinate system is accelerating relative to inertial space.

As a very important special case, consider a coordinate system (x', y', z') rigidly connected to the rotating Earth. Suppose the origin of this coordinate system is at the center of the Earth, as shown in [Figure 15.2](#). The Earth and the coordinate system rotate together about the z' -axis at an angular velocity $\boldsymbol{\Omega}$ (assumed constant). [Figure 15.2](#) also shows an inertial coordinate system (x, y, z) that is at rest with respect to the fixed stars. (I am ignoring the motion of the Earth around the Sun and the motion of the Sun around the center of the Galaxy.)

The common origin of those two coordinate systems is the center of the Earth. The z - and z' -axes coincide and lie along the axis of rotation of the Earth. (You can imagine these axes pointing from the center of the Earth out through the North Pole.) The x, y and x', y' axes all lie in the...